

PASCALX DIGITAL INSTITUTE

60-Hour Cybersecurity SOC Analyst Course

Comprehensive Career Track Brochure | From Security Fundamentals to Hands-On SIEM & Incident Response

 60 Total Hours

 50% Hands-On Labs

 L1 SOC Job Ready

About PascalX

VISION & MISSION

PascalX combines *Pascal* (logical structure and foundational rigor) with *X* (exponential growth and tech transformation).

Our Vision

To be the premier digital institute that empowers the next generation of global tech leaders by replacing passive theory with immersive, lab-first execution—building a skilled workforce capable of securing and innovating the digital future.

Our Mission

To deliver accessible, mentor-led, and career-focused technical education through cutting-edge virtual environments and industry-aligned curricula, taking learners from core fundamentals to job-ready mastery.

 **Lab-First Rigor:** 50%+ Hands-On Labs

 **Expert Mentorship:** Active Industry Practitioners

 **Career Transformation:** Job-Ready Skills

Learning Methodology

CURRICULUM MATRIX

Module Topic	Theory	Practical / Labs	Total Hours	Primary Focus & Outcome
1. Security & SOC Fundamentals	4 Hours	2 Hours	6 Hours	CIA Triad, SOC Workflow, Phishing Analysis
2. Networking & Wireshark	5 Hours	5 Hours	10 Hours	OSI, TCP/IP, Wireshark PCAP, Nmap Scans
3. Windows/Linux Architecture	5 Hours	5 Hours	10 Hours	Event IDs, Sysmon Telemetry, Linux Logs
4. Threat Intel & MITRE ATT&CK	3 Hours	3 Hours	6 Hours	Cyber Kill Chain, TTPs, Open Source OSINT
5. SIEM Mastery (Splunk / ELK)	6 Hours	8 Hours	14 Hours	SPL Queries, Correlation Rules, Alerting
6. Incident Response & Triage	4 Hours	4 Hours	8 Hours	NIST Lifecycle, Ticketing, EDR Basics
7. Capstone & Interview Prep	3 Hours	3 Hours	6 Hours	Live Breach Simulation & Technical Mocks

Program Overview

Designed specifically for beginners and aspiring security professionals, this intensive 60-hour career track bridges the gap between theoretical security concepts and real-world Security Operations Center (SOC) practices.

- ✓ **Balanced Learning:** 30 Hours Theory + 30 Hours Practical Labs
- ✓ **Industry-Standard Tools:** Splunk, Wireshark, Sysmon, Nmap
- ✓ **Scenario-Based Learning:** Real incident investigation workflows
- ✓ **Career Centric:** Resume building, LinkedIn optimization & mock interviews



Modules 1 & 2

FOUNDATIONS & NETWORKS

Mod 1: Intro to SOC (6 Hrs)

- CIA Triad & AAA Security Framework
- SOC Roles: L1, L2, L3 Analyst Workflows
- Threat Landscape: Malware, Ransomware & Phishing
- **Lab:** Phishing email header analysis with VirusTotal

Mod 2: Networking & Protocols (10 Hrs)

- OSI Model & TCP/IP Protocol Suite
- DNS, DHCP, HTTP/S, ARP & Subnetting
- Firewalls, Proxies, VPNs & IDS/IPS
- **Lab:** Deep packet inspection using Wireshark & Nmap scans

Modules 3 & 4

SYSTEMS & THREAT INTEL

Mod 3: OS Artifacts & Telemetry (10 Hrs)

- Windows Architecture, Registry & Active Directory
- Crucial Event IDs (4624, 4625, 4688, 1102)
- Linux Log Hunting (/var/log/auth.log)
- **Lab:** Investigating Sysmon logs for process execution

Mod 4: Threat Intel & Frameworks (6 Hrs)

- Cyber Kill Chain 7-Stage Attack Analysis
- MITRE ATT&CK Tactics, Techniques & Procedures
- Indicators of Compromise (IoCs) vs. IoAs
- **Lab:** OSINT investigation via AlienVault OTX & AbuseIPDB

Module 5

SIEM MASTERY (14 HOURS)



SIEM Fundamentals

Log Collection, Aggregation, Normalization, Correlation Rules, and Parsing telemetry across cloud & enterprise endpoints.



Splunk Search Processing

Hands-on SPL (Search Processing Language) querying, filtering, statistical charting, and building real-time dashboards.



Practical Attack Detection

8 Hours of Practical Labs: Detecting Brute Force attacks, unauthorized domain admin additions, and data exfiltration.

Modules 6 & 7

INCIDENT RESPONSE & CAPSTONE

Mod 6: Incident Response (8 Hrs)

- NIST SP 800-61 Incident Handling Lifecycle
- Alert Triage, Severity Ranking & Escalation
- Endpoint Detection & Response (EDR) basics
- **Lab:** SQL Injection alert triage & incident reporting

Mod 7: Capstone & Career (6 Hrs)

- **Live Breach Capstone:** Full attack investigation
- Ticketing report generation & containment advice
- Resume & LinkedIn optimization for SOC L1 roles
- Mock Technical Interviews (Top 20 SOC questions)

Hands-On Tools

50% LAB FOCUSED



Splunk & ELK

Enterprise SIEM platforms for log correlation, threat hunting, and custom rule development.



Wireshark & Nmap

Industry standard packet analyzer and network discovery tools for deep protocol inspection.



Sysmon & Event Viewer

Windows telemetry logging and process auditing tools to spot malicious activity.

Career Support

JOB READINESS

-  **Portfolio Development:** Complete capstone reports to showcase directly to hiring managers.
-  **1-on-1 Mock Interviews:** Rigorous practice covering technical log questions, OSI model, and scenario triaging.
-  **LinkedIn & Resume Audit:** Tailored keyword optimization for L1 SOC Analyst & Junior Security Operations job postings.
-  **Course Certificate:** Verified PascalX SOC Analyst Certificate of Completion.



Ready to Launch Your Cyber Career with PascalX?

Enroll in the 60-Hour Cybersecurity SOC Analyst Program Today.



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Image Sources



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